

Department of Artificial Intelligence, SVNIT,SURAT
B.Tech-III ,SEM-V
Subject- Machine Learning(AI301)

LAB ASSIGNMENT-6

Problem statement: Dataset: Abalone data (abalone.csv)

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1. Read the data.
2. Do the exploratory data analysis (EDA)
3. Standardize the input
4. Build the Logistic Regression model, then make a prediction on the test dataset.
5. Print the accuracy_score, confusion_matrix, classification_report for Logistic Regression
6. model
7. Perform PCA, then use print (pca_model.explained_variance_ratio_) to significant variables
8. Visualize coefficients with a heat map using the Principal components
9. Now with Principal components, build and test Logistic Regression and Support Vector
10. Classification
11. Print the accuracy_score, confusion_matrix, classification_report for built model for previous step